# Derivative bounds in the velocity variable for the linear and nonlinear Wigner terms

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*Editorial note.* The counted statement is the author's labelled lemma 'etalemma' (source lines 715-728); the proof environment begins on the next line after \end{lemma}, so the pairing is by adjacency.

*Editorial note.* The retained context is the author's own lead-in, including the display that defines the terms L_P and N_P and the remark about the order O(hbar) terms that the statement's second summand encodes.

*Editorial note.* No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

*Editorial note.* No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference; the retained preamble is the publisher layout adapter plus the native macro, operator and theorem block of the same pinned source.

*Editorial scope.* One original result of the pinned publisher article, counted once: the velocity-derivative bounds for the linear and nonlinear Wigner terms (author label etalemma).

*Allowed standard prerequisites (editorial).*

- The linear and nonlinear Wigner terms. L_P and N_P are the two terms of the Wigner equation, defined by the displays preceding the lemma, with L_P carrying the oscillatory factor sin((hbar/2) k.(eta - k t)).. Both bounds in the statement are about the derivatives in eta of the Wigner transforms of L_P and N_P.
- Leibniz rule for multi-index derivatives. D^alpha of a product expands as a sum over beta <= alpha with multinomial coefficients.. It is the first step of the proof at source lines 730-737.
- Elementary trigonometric bounds. |sin x| <= |x| and |sin x|, |cos x| <= 1.. They separate the beta = alpha term from the beta < alpha terms at source lines 738-745 and produce the two different powers of hbar |k|.

Essential to the analysis are bounds on derivatives of the Wigner transform in the $\eta$ variable, $D_\eta^\alpha \widehat{W[P]}$, corresponding to velocity weights $\boldsymbol{\xi}$ on $P$. For multi-indices $\alpha$ with $|\alpha| \geq 2$, additional terms of order $O(\hbar)$ arise from differentiation of the oscillatory phase, in stark contrast to the classical case.

**Lemma 0.1.** *Let $\alpha$ be a multi-index. The Wigner transform of $\mathrm{L}_P$ satisfies*

$$\begin{aligned}\big|D_\eta^\alpha \widehat{W[\mathrm{L}_P]}(t,k,\eta)\big| \lesssim |k||\widehat{w}(k)||\widehat{\rho_Q}(t,k)||\eta-kt|\big|D_\eta^\alpha \widehat{g}(\eta-kt)\big| \\ + |k|\langle \hbar k\rangle^{|\alpha|-1}|\widehat{w}(k)||\widehat{\rho_Q}(t,k)| \sum_{|\beta|\le|\alpha|-1}\big|D_\eta^\beta \widehat{g}(\eta-kt)\big|.\end{aligned}$$

*The Wigner transform of $\mathrm{N}_P$ satisfies*

$$\begin{aligned}&\big|D_\eta^\alpha \widehat{W[\mathrm{N}_P]}(t,k,\eta)\big| \\ &\qquad\lesssim \int_\ell |\ell|\,|\widehat{w}(\ell)|\,|\widehat{\rho_Q}(t,\ell)|\,|\eta-kt|\,\big|D_\eta^\alpha \widehat{W[P]}(t,k-\ell,\eta-\ell t)\big|\,d\ell \\ &\qquad+\sum_{|\beta|\le|\alpha|-1}\int_\ell |\ell|\,\langle\hbar\ell\rangle^{|\alpha|-1}\,|\widehat{w}(\ell)|\,|\widehat{\rho_Q}(t,\ell)|\,\big|D_\eta^\beta \widehat{W[P]}(t,k-\ell,\eta-\ell t)\big|\,d\ell.\end{aligned}$$

*When $|\alpha|=0$, sums over $|\beta|\le|\alpha|-1$ are understood to be empty.*

*Proof.* Applying the Leibniz rule to the linear term gives

$$D_\eta^\alpha\left[\sin\left(\tfrac{\hbar}{2}k\cdot(\eta-kt)\right)\widehat{g}(\eta-kt)\right]=\sum_{\beta\le\alpha}\binom{\alpha}{\beta}D_\eta^{\alpha-\beta}\left[\sin\left(\tfrac{\hbar}{2}k\cdot(\eta-kt)\right)\right]D_\eta^\beta\widehat{g}(\eta-kt).$$

For the term with $\beta=\alpha$, use $|\sin x|\le|x|$ to get

$$\left|\sin\left(\tfrac{\hbar}{2}k\cdot(\eta-kt)\right)\right|\le\hbar|k||\eta-kt|.$$

For terms with $\beta<\alpha$, differentiate sin directly and use $|\sin x|,|\cos x|\le 1$ to obtain

$$\left|D_\eta^{\alpha-\beta}\left[\sin\left(\tfrac{\hbar}{2}k\cdot(\eta-kt)\right)\right]\right|\le(\hbar|k|)^{|\alpha-\beta|}.$$

This leads to the estimate

$$\begin{aligned}&\left|D_\eta^\alpha\left[\left(e^{-i(\hbar/2)k\cdot(\eta-kt)}-e^{i(\hbar/2)k\cdot(\eta-kt)}\right)\widehat{g}(\eta-kt)\right]\right| \\ &\qquad\lesssim\hbar|k||\eta-kt|\big|D_\eta^\alpha\widehat{g}(\eta-kt)\big|+\sum_{\beta<\alpha}(\hbar|k|)^{|\alpha-\beta|}\big|D_\eta^\beta\widehat{g}(\eta-kt)\big|.\end{aligned}$$

Since for any $\beta<\alpha$, $(\hbar|k|)^{|\alpha-\beta|}\lesssim\hbar|k|\langle\hbar k\rangle^{|\alpha|-1}$, this gives the bound for $\widehat{W[\mathrm{L}_P]}$. The estimate for $\widehat{W[\mathrm{N}_P]}$ follows similarly. □

# References


[1] J. Bedrossian - "Nonlinear echoes and Landau damping with insufficient regularity", Tunis. J. Math. 3 (2021) no. 1, p. 121-205

[2] J. Bedrossian, N. Masmoudi & C. Mouhot - "Landau damping: paraproducts and Gevrey regularity", Ann. PDE 2 (2016) no. 1, article ID 4, 71 pages

[3] J. Bedrossian, N. Masmoudi & C. Mouhot - "Landau damping in finite regularity for unconfined systems with screened interactions", Comm. Pure Appl. Math. 71 (2018) no. 3, p. 537-576

[4] J. Bedrossian, N. Masmoudi & C. Mouhot - "Linearized wave-damping structure of Vlasov–Poisson in $\mathbb{R}^3$", SIAM J. Math. Anal. 54 (2022) no. 4, p. 4379-4406

[5] A. Borie, S. Hadama & J. Sabin - "Scattering for the positive density Hartree equation", 2025

[6] A. Bove, G. Da Prato & G. Fano - "An existence proof for the Hartree-Fock time-dependent problem with bounded two-body interaction", Comm. Math. Phys. 37 (1974), p. 183-191

[7] A. Bove, G. Da Prato & G. Fano - "On the Hartree-Fock time-dependent problem", Comm. Math. Phys. 49 (1976) no. 1, p. 25-33

[8] T. Cazenave - Semilinear Schrödinger equations, Courant Lecture Notes in Math., vol. 10, American Mathematical Society, Providence, RI, 2003

[9] J. M. Chadam - "The time-dependent Hartree-Fock equations with Coulomb two-body interaction", Comm. Math. Phys. 46 (1976) no. 2, p. 99-104

[10] T. Chen, Y. Hong & N. Pavlović - "Global well-posedness of the NLS system for infinitely many fermions", Arch. Rational Mech. Anal. 224 (2017) no. 1, p. 91-123

[11] T. Chen, Y. Hong & N. Pavlović - "On the scattering problem for infinitely many fermions in dimensions $d \geq 3$ at positive temperature", Ann. Inst. H. Poincaré C Anal. Non Linéaire 35 (2018) no. 2, p. 393-416

[12] J. J. Chong, L. Lafleche & C. Saffirio - "On the $L^2$ rate of convergence in the limit from the Hartree to the Vlasov-Poisson equation", J. Éc. polytech. Math. 10 (2023), p. 703-726

[13] C. Collot & A.-S. de Suzzoni - "Stability of equilibria for a Hartree equation for random fields", J. Math. Pures Appl. (9) 137 (2020), p. 70-100

[14] C. Collot & A.-S. de Suzzoni - "Stability of steady states for Hartree and Schrödinger equations for infinitely many particles", Ann. H. Lebesgue 5 (2022), p. 429-490

[15] E. Grenier, T. T. Nguyen & I. Rodnianski - "Landau damping for analytic and Gevrey data", Math. Res. Lett. 28 (2021) no. 6, p. 1679-1702

[16] S. Hadama - "Asymptotic stability of a wide class of stationary solutions for the Hartree and Schrödinger equations for infinitely many particles", Ann. H. Lebesgue 8 (2025), p. 181-218

[17] A. D. Ionescu, B. Pausader, X. Wang & K. Widmayer - "Nonlinear Landau damping for the Vlasov-Poisson system in $\mathbb{R}^3$: the Poisson equilibrium", Ann. PDE 10 (2024) no. 1, article ID 2, 78 pages

[18] L. Landau - "On the vibrations of the electronic plasma", Acad. Sci. USSR. J. Phys. 10 (1946), p. 25-34

[19] M. Lewin & J. Sabin - "The Hartree equation for infinitely many particles, II: Dispersion and scattering in 2D", Anal. PDE 7 (2014) no. 6, p. 1339-1363

[20] M. Lewin & J. Sabin - "The Hartree equation for infinitely many particles I. Well-posedness theory", Comm. Math. Phys. 334 (2015) no. 1, p. 117-170

[21] M. Lewin & J. Sabin - "The Hartree and Vlasov equations at positive density", Comm. Partial Differential Equations 45 (2020) no. 12, p. 1702-1754

[22] C. Mouhot & C. Villani - "On Landau damping", Acta Math. 207 (2011) no. 1, p. 29-201

[23] T. T. Nguyen & C. You - "Modified scattering for long-range Hartree equations of infinite rank near vacuum", SIAM J. Math. Anal. 57 (2025) no. 6, p. 6486-6497

[24] T. T. Nguyen & C. You - "Plasmons for the Hartree equations with Coulomb interaction", Probab. Math. Phys. 6 (2025) no. 3, p. 913-960

[25] E. M. Stein & G. Weiss - "Interpolation of operators with change of measures", Trans. Amer. Math. Soc. 87 (1958), p. 159-172

[26] C. You - "Phase mixing estimates for the nonlinear Hartree equation of infinite rank", 2024